In [1]:
```
import os
print(os.getcwd())
```

```
C:\Users\DELL
```

In [7]:
```
import pandas as pd
df = pd.read_csv(r"C:\Users\DELL\Downloads\customer_shopping_behavior.csv")
```

In [8]:
```
df.head()
```

Out[8]:

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise

In [9]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Customer ID             3900 non-null   int64
 1   Age                     3900 non-null   int64
 2   Gender                  3900 non-null   str
 3   Item Purchased          3900 non-null   str
 4   Category                3900 non-null   str
 5   Purchase Amount (USD)   3900 non-null   int64
 6   Location                3900 non-null   str
 7   Size                    3900 non-null   str
 8   Color                   3900 non-null   str
 9   Season                  3900 non-null   str
 10  Review Rating           3863 non-null   float64
 11  Subscription Status     3900 non-null   str
 12  Shipping Type           3900 non-null   str
 13  Discount Applied        3900 non-null   str
 14  Promo Code Used         3900 non-null   str
 15  Previous Purchases      3900 non-null   int64
 16  Payment Method          3900 non-null   str
 17  Frequency of Purchases  3900 non-null   str
dtypes: float64(1), int64(4), str(13)
memory usage: 548.6 KB
```

In [10]:
```
df.describe(include='all')
```

Out[10]:

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	S
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	39
unique	NaN	NaN	2	25	4	NaN	50	
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	
freq	NaN	NaN	2652	171	1737	NaN	96	17
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	N
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	N
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	N
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	N
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	N
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	N
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	N

In [11]:
```
df.isnull().sum()
```

Out[11]:
```
Customer ID                0
Age                        0
Gender                     0
Item Purchased             0
Category                   0
Purchase Amount (USD)      0
Location                   0
Size                       0
Color                      0
Season                     0
Review Rating             37
Subscription Status        0
Shipping Type              0
Discount Applied           0
Promo Code Used            0
Previous Purchases         0
Payment Method             0
Frequency of Purchases     0
dtype: int64
```

In [12]:
```
df['Review Rating'] = df.groupby('Category')['Review Rating'].transform(lambda x: x
```

In [13]:
```
df.isnull().sum()
```

Out[13]:
```
Customer ID                0
Age                        0
Gender                     0
Item Purchased             0
Category                   0
Purchase Amount (USD)      0
Location                   0
Size                       0
Color                      0
Season                     0
Review Rating              0
Subscription Status        0
Shipping Type              0
Discount Applied           0
Promo Code Used            0
Previous Purchases         0
Payment Method             0
Frequency of Purchases     0
dtype: int64
```

In [14]:
```
df.columns = df.columns.str.lower()
df.columns = df.columns.str.replace(' ','_')
df=df.rename(columns={'purchase_amount_(usd)':'purchase_amount'})
```

In [15]:
```
df.columns
```

Out[15]:
```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
       'purchase_amount', 'location', 'size', 'color', 'season',
       'review_rating', 'subscription_status', 'shipping_type',
       'discount_applied', 'promo_code_used', 'previous_purchases',
       'payment_method', 'frequency_of_purchases'],
      dtype='str')
```

In [16]:
```
#creating a column age_group
labels=['Young Adult', 'Adult', 'Middle-aged', 'Senior']
df['age_group']=pd.qcut(df['age'],q=4, labels=labels)
```

In [17]:
```
df[['age','age_group']].head(10)
```

Out[17]:

	age	age_group
0	55	Middle-aged
1	19	Young Adult
2	50	Middle-aged
3	21	Young Adult
4	45	Middle-aged
5	46	Middle-aged
6	63	Senior
7	27	Young Adult
8	26	Young Adult
9	57	Middle-aged

In [18]:
```
#purchase_freq days creation
frequency_mapping={
    'Fortnightly':14,
    'Weekly':7,
    'Monthly':30,
    'Quarterly':90,
    'Bi-Weekly':14,
    'Annually':365,
    'Every 3 Months':90
}
df['purchase_frequency_days']=df['frequency_of_purchases'].map(frequency_mapping)
```

In [19]:
```
df[['purchase_frequency_days','frequency_of_purchases']].head(10)
```

Out[19]:

	purchase_frequency_days	frequency_of_purchases
0	14	Fortnightly
1	14	Fortnightly
2	7	Weekly
3	7	Weekly
4	365	Annually
5	7	Weekly
6	90	Quarterly
7	7	Weekly
8	365	Annually
9	90	Quarterly

In [20]:
```
df[['discount_applied','promo_code_used']].head(10)
```

Out[20]:

	discount_applied	promo_code_used
0	Yes	Yes
1	Yes	Yes
2	Yes	Yes
3	Yes	Yes
4	Yes	Yes
5	Yes	Yes
6	Yes	Yes
7	Yes	Yes
8	Yes	Yes
9	Yes	Yes

In [21]:
```
(df['discount_applied'] == df['promo_code_used']).all()
```

Out[21]:
```
np.True_
```

In [22]:
```
df=df.drop('promo_code_used', axis=1)
```

In [23]:
```
df.columns
```

Out[23]:
```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
       'purchase_amount', 'location', 'size', 'color', 'season',
       'review_rating', 'subscription_status', 'shipping_type',
       'discount_applied', 'previous_purchases', 'payment_method',
       'frequency_of_purchases', 'age_group', 'purchase_frequency_days'],
      dtype='str')
```

In [24]:
```
!pip install psycopg2-binary sqlalchemy
```

```
Collecting psycopg2-binary
  Downloading psycopg2_binary-2.9.12-cp313-cp313-win_amd64.whl.metadata (5.1 kB)
Collecting sqlalchemy
  Downloading sqlalchemy-2.0.49-cp313-cp313-win_amd64.whl.metadata (9.8 kB)
Collecting greenlet>=1 (from sqlalchemy)
  Downloading greenlet-3.4.0-cp313-cp313-win_amd64.whl.metadata (3.8 kB)
Requirement already satisfied: typing-extensions>=4.6.0 in .\miniconda3\Lib\site-pac
kages (from sqlalchemy) (4.15.0)
Downloading psycopg2_binary-2.9.12-cp313-cp313-win_amd64.whl (2.8 MB)
   ---------------------------------------- 0.0/2.8 MB ? eta -:--:--
   ------------------------------ --------- 2.1/2.8 MB 11.0 MB/s eta 0:00:01
   ---------------------------------------- 2.8/2.8 MB 9.7 MB/s  0:00:00
Downloading sqlalchemy-2.0.49-cp313-cp313-win_amd64.whl (2.1 MB)
   ---------------------------------------- 0.0/2.1 MB ? eta -:--:--
   ---------------------------------------  2.1/2.1 MB 10.9 MB/s eta 0:00:01
   ---------------------------------------- 2.1/2.1 MB 9.7 MB/s  0:00:00
Downloading greenlet-3.4.0-cp313-cp313-win_amd64.whl (238 kB)
Installing collected packages: psycopg2-binary, greenlet, sqlalchemy

   ---------------------------------------- 0/3 [psycopg2-binary]
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   ---------------------------------------- 3/3 [sqlalchemy]

Successfully installed greenlet-3.4.0 psycopg2-binary-2.9.12 sqlalchemy-2.0.49
Note: you may need to restart the kernel to use updated packages.
```

In [25]:

```python
from sqlalchemy import create_engine

username = "postgres"
password = "uday12"
host = "localhost"
port = "5432"
database = "customer_behavior"

engine = create_engine(
    f"postgresql+psycopg2://{username}:{password}@{host}:{port}/{database}"
)

table_name = "customer"

df.to_sql(table_name, engine, if_exists="replace", index=False)

print(f"Data successfully loaded into table '{table_name}' in database '{database}'
```

```
Data successfully loaded into table 'customer' in database 'customer_behavior'
```

In [ ]: